WIDS 2024: Facial Recognition - Project Report

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1 Basics

- Learnt to use basic Python modules such as **NumPy**, **Matplotlib** and **Pandas**
- Built a **Linear Regression** model to predict the marks of students based on their habits, and used **Gradient Descent** to train the model

2 Main Project

- Imported **OpenCV** library for image processing, and **TensorFlow** library to create the Neural Network model
- Used the Labelled Faces in the Wild **(LFW)** dataset and **800 scanned images** of myself to train the model
- Created embedding layer of the **Siamese Neural Network** as four **Convulation** layers with **ReLU** activation and finally a **sigmoid** activation layer for output
- Used the **absolute** function for the distance layer and used the **sigmoid** function for the final classification
- Trained the model with **Binary Cross Entropy** as the loss function for **50 epochs**
- Evaluated the model to get a **Precision** of **1** and **Recall** of **1**